Slide 8 – Challenges We Overcame

Introduction

During development, we faced four major technical challenges. We identified the cause of each problem and fixed it, making our application stable and reliable.

1. Database Connection Pooling

Problem: The Spring Boot backend could not connect properly to the Supabase database.
Solution: We used the Supabase Transaction Pooler, which manages database connections and made the connection stable.

2. Prepared Statement Conflict

Problem: Our application was giving 500 Internal Server Errors while running database queries.
Solution: We changed the database configuration (prepareThreshold=0) so Spring Boot could communicate with the database correctly.

3. CORS (Same-Origin Policy)

Problem: The browser blocked communication because the frontend and backend were hosted on different websites.
Solution: We hosted both the frontend and backend together in Spring Boot, so the browser allowed the requests.

4. MultipartFile Stream Closed

Problem: The uploaded PDF became unavailable during background processing.
Solution: We extracted the resume text first using Apache PDFBox, then started the asynchronous process.

Slide 9 – Cloud Infrastructure

Introduction

This slide explains how our project is deployed online using three cloud services: Railway, Supabase, and GitHub.

1. Railway (Backend + CI/CD)

What is it? Railway hosts our Spring Boot application on the cloud.
What does it do? It automatically deploys the latest version whenever we push new code to GitHub and makes the application available online.

2. Supabase (PostgreSQL Database)

What is it? Supabase provides a cloud PostgreSQL database.
What does it do? It stores developer scores, analysis results, and processing status safely.

3. GitHub (Source Code + API)

What is it? GitHub stores our project source code and also provides developer data through the GitHub REST API.
What does it do? Railway gets the latest code from GitHub, while our application fetches repositories, commits, languages, and profile details from the GitHub API.

How They Work Together

The user uploads a resume and GitHub username. Railway runs the application, GitHub provides developer data, Supabase stores the results, and finally the application displays the developer score.